

Counterexamples to the proposed joint coreset bound at $p = 4$

Explicit constructions, a sharper probabilistic lower bound,
and reproducible verification

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Abstract

We give a negative answer, already at $p = 4$, to the displayed size bound in RA-05. For $r = 4^h$, there is an explicit matrix in $2r$ dimensions such that, when $k = 2r - 1$ and $\varepsilon = \Theta(r^{-3/4})$, every strong row coreset has $\Omega(r^3)$ rows. The proposed expression is only $O(r^{11/4} \log^c r)$ for any fixed logarithmic exponent c . The construction even excludes signed reweightings of the same support size. A second explicit construction yields $\Omega(k^2/(\varepsilon^2 + k^{-1/2}))$ rows. A probabilistic strengthening gives $\Omega(k^2/(\varepsilon^2 + (\log k)/k))$ rows, matching the cited $\tilde{O}(k^2 \varepsilon^{-2})$ upper bound when $\varepsilon \geq \sqrt{(\log k)/k}$ and $\varepsilon \leq 1/4$. All constructions and lower-bound arguments are proved below; the finite-field construction underlying the explicit examples is included.

Scope and status. This manuscript refutes the proposed bound at $p = 4$ and identifies the optimal order up to logarithms in a stated $p = 4$ accuracy regime. It does *not* determine the optimal joint function for every fixed $p > 2$ and every accuracy. It is an original argument prepared in this session, not a claim of peer-reviewed priority or a machine-checked formal proof. Numerical tests are supplementary; the general statements rest on the proofs.

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1 Problem, results, and comparison with the proposed bound

1.1 The row-subset model

For a matrix $A \in \mathbb{R}^{n \times d}$, let a_i denote its rows written as column vectors. The quartic subspace cost is

$$\text{cost}_A(F) = \sum_{i=1}^n \|(I - P_F)a_i\|_2^4. \quad (1)$$

A strong row coreset uses nonnegative weights w_i and preserves this cost within a multiplicative factor $1 \pm \varepsilon$ for every linear subspace of dimension at most k . Its size is the number of indices with $w_i \neq 0$, not the sum of the weights. Define $S_4(k, \varepsilon)$ as the worst-case minimum support over all n, d, A with $k < d$.

RA-05 asks for the optimal joint dependence and, in particular, whether

$$C_p \left(\frac{k^{p/2}}{\varepsilon} + \frac{k}{\varepsilon^2} \right) \log^{c_p} \left(\frac{2k}{\varepsilon} \right) \quad (2)$$

is an upper bound with constants depending only on fixed $p > 2$ [1]. The cited upper bound is $\tilde{O}_p(k^{p/2}\varepsilon^{-2})$; Lin–Mirrokni–Woodruff’s Theorem 1.2 provides it, while their Section 1.4 records the joint-dependence gap [2]. We use that upper bound only when interpreting our lower bounds. None of our lower-bound proofs depend on it.

1.2 An explicit counterexample

Throughout, put

$$r = 4^h \ (h \geq 1), \quad B = \frac{r}{2} + 1, \quad M = Br = \frac{r(r+2)}{2}, \quad \lambda = 1 - \frac{1}{r}. \quad (3)$$

Let $u_1, \dots, u_M \in \mathbb{R}^r$ be the union of B real mutually unbiased orthonormal bases. This means that vectors from different bases have inner product of absolute value $r^{-1/2}$. Appendix A constructs these bases explicitly.

Theorem 1.1 (Explicit orthogonal-noise counterexample). *Set $\delta = B^{-1/4}$ and form the Mr rows*

$$a_{jt} = r^{-1/4}(u_j, \delta e_t) \in \mathbb{R}^{2r}, \quad 1 \leq j \leq M, \quad 1 \leq t \leq r. \quad (4)$$

If real weights on these rows preserve all hyperplane costs within $1 \pm \varepsilon$, their support size m satisfies

$$m \geq Mr \left(1 - \frac{100r\varepsilon^2}{\lambda^2\delta^2} \right). \quad (5)$$

In particular, for

$$k = 2r - 1, \quad \varepsilon_r = \frac{\delta}{20\sqrt{r}}, \quad (6)$$

every strong row coreset has $m \geq (5/9)Mr = \Omega(r^3)$ rows.

Since $\varepsilon_r = \Theta(r^{-3/4})$, the two polynomial terms in (2), at $p = 4$, are respectively $\Theta(r^{11/4})$ and $\Theta(r^{5/2})$. Also $\log(2k/\varepsilon_r) = \Theta(\log r)$. Consequently

$$\frac{r^3}{(r^{11/4} + r^{5/2})(\log r)^c} \longrightarrow \infty \quad \text{for every fixed } c > 0.$$

Thus no constants C_4, c_4 make (2) a universal bound. One counterexample exponent $p = 4$ suffices to disprove the proposed formula as an assertion for every fixed $p > 2$; it does not classify the other exponents.

1.3 Stronger bounds over ranges of accuracies

Theorem 1.2 (Explicit product construction). *For each r in (3), there is an explicit M^2 -row matrix in $2r$ dimensions such that every nonnegative strong row coreset, for $k = 2r - 1$ and $0 < \varepsilon \leq 1/4$, has*

$$m \geq \frac{M}{1568\varepsilon^2 + 13r^{-1/2}}. \quad (7)$$

In particular, at $\varepsilon = r^{-1/4}/4$, it requires $m \geq M\sqrt{r}/111 = \Omega(r^{5/2})$ rows.

At the latter accuracy the proposed expression is only $O(r^{9/4} \log^c r)$, giving a second polynomial separation. More generally, (7) matches the cited upper bound, up to logarithms, for $r^{-1/4} \leq \varepsilon \leq 1/4$. The following strengthening replaces the coherence argument used there by a sparse singular-value argument.

Theorem 1.3 (Probabilistic strengthening). *For every $r = 4^h \geq 4^7$, there is a matrix in $2r$ dimensions for which any real reweighting preserving all hyperplane costs at accuracy $0 < \varepsilon \leq 1/4$ has*

$$m \geq \frac{M}{492032\varepsilon^2 + 11520(\log r)/r}. \quad (8)$$

The matrix has $100Mr^3$ rows. Its existence is proved by a positive-probability construction, with every needed event estimated explicitly.

Zero-padding, detailed in Section 6, implies that for all sufficiently large integers k and $0 < \varepsilon \leq 1/4$,

$$S_4(k, \varepsilon) \geq c \frac{k^2}{\varepsilon^2 + (\log k)/k} \quad (9)$$

for an absolute constant $c > 0$. Combining this with the cited upper bound gives

$$S_4(k, \varepsilon) = \tilde{\Theta}(k^2 \varepsilon^{-2}) \quad \text{when} \quad \sqrt{\frac{\log k}{k}} \leq \varepsilon \leq \frac{1}{4}. \quad (10)$$

The stronger bound is not needed to refute (2); Theorem 1.1 already does so by an entirely explicit construction.

2 Hyperplanes and the algebra of the core vectors

2.1 Reduction to quartic forms

Lemma 2.1 (Hyperplane reduction). *Let $d = k + 1$. A strong row coreset at rank k necessarily satisfies*

$$\left| \sum_i (w_i - 1)(a_i \cdot z)^4 \right| \leq \varepsilon \sum_i (a_i \cdot z)^4 \quad \text{for every } z \in \mathbb{R}^d. \quad (11)$$

Proof. For a unit vector z , the hyperplane $F = z^\perp$ has dimension k , and $I - P_F = zz^T$. Hence $\|(I - P_F)a_i\|_2^4 = (a_i \cdot z)^4$. The strong guarantee gives (11) on the unit sphere. Both sides are homogeneous of degree four, so it holds for every nonzero z , and at zero it is trivial. $\square$

We only use this necessary condition. There is no need to assert that hyperplane preservation suffices for preservation of smaller-dimensional subspaces.

2.2 Moment and kernel identities

Write $U \in \mathbb{R}^{M \times r}$ for the matrix with rows u_j^T , and $G = UU^T$. An entrywise power, such as $G^{\circ 3}$, is not a matrix power.

Lemma 2.2 (Complete real MUB identities). *The core vectors satisfy*

$$U^T U = B I_r, \quad (12)$$

$$\sum_{j=1}^M (u_j \cdot x)^4 = \frac{3}{2} \|x\|_2^4, \quad x \in \mathbb{R}^r, \quad (13)$$

$$K := G^{\circ 3} = \left(1 - \frac{1}{r}\right) I_M + \frac{1}{r} G, \quad K \succeq \lambda I_M, \quad (14)$$

$$L := G^{\circ 4} \succeq \lambda I_M. \quad (15)$$

Proof. Equation (12) follows by adding the orthonormal-basis identities. For (13), index vectors as u_{bt} by basis and set $Q_{bt} = u_{bt} u_{bt}^T - I_r/r$. For each b , the matrices Q_{bt} span the $(r-1)$ -dimensional space of traceless matrices diagonal in that basis. For $b \neq c$,

$$\langle Q_{bt}, Q_{cs} \rangle_F = (u_{bt} \cdot u_{cs})^2 - \frac{1}{r} = 0.$$

These spaces are pairwise orthogonal, and their total dimension is

$$B(r-1) = \frac{r(r+1)}{2} - 1,$$

the dimension of the space of all real symmetric traceless matrices. For any symmetric traceless X , squared projection lengths therefore sum to

$$\sum_{b,t} \langle X, Q_{bt} \rangle_F^2 = \|X\|_F^2.$$

Within a fixed basis this is simply the sum of squares of the diagonal entries of X ; their sum is zero. Take $X = xx^T - \|x\|^2 I_r/r$ and expand:

$$\sum_j (u_j \cdot x)^4 - \frac{B}{r} \|x\|^4 = \left(1 - \frac{1}{r}\right) \|x\|^4.$$

Since $B/r = 1/2 + 1/r$, (13) follows.

The diagonal entries of G are one; its off-diagonal entries are either zero or $\pm r^{-1/2}$. Entrywise cubing gives (14). For L , diagonal basis blocks are I_r and off-diagonal basis blocks are $r^{-2} J_r$. The subspace of vectors summing to zero within every basis has eigenvalue one. On vectors constant on each basis, the induced matrix is

$$I_B + \frac{1}{r} (J_B - I_B),$$

with eigenvalues $1 - 1/r$ and $1 + (B-1)/r = 3/2$. This proves (15). $\square$

In particular, for every real matrix H with M rows,

$$\|KH\|_F \geq \lambda \|H\|_F, \quad \|LH\|_F \geq \lambda \|H\|_F. \quad (16)$$

This conditioning is what prevents reweighting unrelated core groups from canceling their errors.

3 Proof of the explicit orthogonal-noise counterexample

3.1 The original cost is uniformly controlled

For the rows in (4), write

$$f(x, y) = \frac{1}{r} \sum_{j=1}^M \sum_{t=1}^r (u_j \cdot x + \delta y_t)^4. \quad (17)$$

The scalar inequality $(a + b)^4 \leq 8(a^4 + b^4)$ and (13) imply

$$\begin{aligned} f(x, y) &\leq 8 \sum_j (u_j \cdot x)^4 + \frac{8M\delta^4}{r} \sum_t y_t^4 \\ &\leq 12 \|x\|^4 + 8 \|y\|^4, \end{aligned} \quad (18)$$

since $M\delta^4/r = 1$ and $\sum y_t^4 \leq \|y\|^4$.

Let f_w denote the weighted version and put

$$g = f_w - f, \quad H_{jt} = \frac{w_{jt} - 1}{r}.$$

By Lemma 2.1, $|g(x, y)| \leq \varepsilon f(x, y)$.

3.2 Four queries isolate the cubic-linear interaction

Define

$$A(x, y) = \sum_{j,t} H_{jt} (u_j \cdot x)^3 y_t, \quad D(x, y) = \sum_{j,t} H_{jt} (u_j \cdot x) y_t^3.$$

The part odd in y is

$$O(x, y) := \frac{g(x, y) - g(x, -y)}{2} = 4\delta A(x, y) + 4\delta^3 D(x, y).$$

Scaling x separates these two terms exactly:

$$O(2x, y) - 2O(x, y) = 24\delta A(x, y). \quad (19)$$

For unit x, y , (18) gives $|O(x, y)| \leq 20\varepsilon$ and $|O(2x, y)| \leq 200\varepsilon$. Consequently

$$|A(x, y)| \leq \frac{10\varepsilon}{\delta}, \quad \left\| \sum_j (u_j \cdot x)^3 H_{j,:} \right\|_2 \leq \frac{10\varepsilon}{\delta}. \quad (20)$$

The second inequality follows by taking the supremum over unit y . For $x = u_i$, its left side is the norm of row i of KH . Thus

$$\|KH\|_F^2 \leq \frac{100M\varepsilon^2}{\delta^2}, \quad \|H\|_F^2 \leq \frac{100M\varepsilon^2}{\lambda^2\delta^2}.$$

Every omitted row has $w_{jt} = 0$, hence $H_{jt} = -1/r$. If there are m retained rows,

$$\frac{Mr - m}{r^2} \leq \|H\|_F^2.$$

Rearranging proves (5). At (6),

$$m \geq Mr \left(1 - \frac{1}{4\lambda^2}\right) \geq \frac{5}{9}Mr,$$

because $r \geq 4$ gives $\lambda \geq 3/4$. No part of this proof assumes nonnegative weights.

3.3 Extracting an actual violating hyperplane

The proof gives more than a support count. For arbitrary candidate weights, compute H and then

$$KH = \left(1 - \frac{1}{r}\right)H + \frac{1}{r}U(U^T H). \quad (21)$$

Choose a row i of maximum norm. If that norm is nonzero, set

$$x = u_i, \quad y = \frac{(KH)_{i,:}}{\|(KH)_{i,:}\|_2}.$$

Among the four query vectors $(x, y), (x, -y), (2x, y), (2x, -y)$, at least one has relative cost error at least

$$\frac{\delta}{10} \|(KH)_{i,:}\|_2. \quad (22)$$

Indeed, applying the preceding argument with the largest of their four relative errors in place of ε gives exactly this inequality. Normalize the selected vector z and return the hyperplane $z^\perp$. Homogeneity leaves its relative error unchanged. Formula (21) avoids constructing an $M \times M$ matrix.

4 Proof of the explicit product lower bound

4.1 Noise vectors and group masses

Choose signs $\sigma_t \in \{-1, 1\}$ and let $v_t = \sigma_t u_t$ so that

$$\mu = \frac{1}{M} \sum_t v_t, \quad \|\mu\|^2 \leq \frac{1}{M}. \quad (23)$$

This is constructive: if S is the running sum, choose the sign of the next vector to make $\langle S, v_t \rangle \leq 0$. Then $\|S + v_t\|^2 \leq \|S\|^2 + 1$, and induction proves the bound. Changing signs preserves (13) and absolute coherences. Use the rows

$$a_{jt} = M^{-1/4}(u_j, v_t), \quad 1 \leq j, t \leq M. \quad (24)$$

Their quartic form satisfies

$$f(x, y) = \frac{1}{M} \sum_{j,t} (u_j \cdot x + v_t \cdot y)^4 \leq 12(\|x\|^4 + \|y\|^4). \quad (25)$$

For nonnegative candidate weights put

$$\alpha_{jt} = \frac{w_{jt}}{M}, \quad s_j = \sum_t \alpha_{jt}, \quad b_j = \sum_t \alpha_{jt} v_t.$$

The original values are $s_j = 1$ and $b_j = \mu$. At $y = 0$, the error is $\sum_j (s_j - 1)(u_j \cdot x)^4$. Taking $x = u_i$ yields

$$\|L(s - \mathbf{1})\|_\infty \leq \frac{3}{2}\varepsilon.$$

Lemma 2.2 gives

$$\|s - \mathbf{1}\|_2 \leq 2\varepsilon\sqrt{M}, \quad \sum_j s_j \geq (1 - 2\varepsilon)M, \quad \sum_j s_j^2 \leq (1 + 2\varepsilon)^2 M. \quad (26)$$

4.2 The mixed interaction controls all group means

Let $g = f_w - f$ and let O again be its part odd in y . The coefficient of degree three in x and degree one in y is

$$A(x, y) = \sum_j (u_j \cdot x)^3 \langle b_j - \mu, y \rangle.$$

The same polynomial extraction gives $O(2x, y) - 2O(x, y) = 24A(x, y)$. For unit x, y , the bounds are now $|O(x, y)| \leq 24\varepsilon$ and $|O(2x, y)| \leq 204\varepsilon$. Hence $|A(x, y)| \leq (21/2)\varepsilon$. Evaluating at $x = u_i$, taking the supremum in y , and using $\lambda \geq 3/4$ gives

$$\sum_j \|b_j - \mu\|^2 \leq 196M\varepsilon^2. \quad (27)$$

Together with (23),

$$\sum_j \|b_j\|^2 \leq 392M\varepsilon^2 + 2. \quad (28)$$

4.3 Positive weights cannot realize these means too sparsely

Set $\nu = r^{-1/2}$ and let q_j be the number of nonzero weights in group j . For $q_j > 0$, coherence and nonnegativity give

$$\begin{aligned} \|b_j\|^2 &\geq (1 + \nu) \sum_t \alpha_{jt}^2 - \nu s_j^2 \\ &\geq \frac{s_j^2}{q_j} - \nu s_j^2. \end{aligned} \quad (29)$$

If $q_j = 0$, then $s_j = b_j = 0$, and its contribution s_j^2/q_j is defined as zero. With $m = \sum_j q_j$, Cauchy–Schwarz implies

$$\sum_j \frac{s_j^2}{q_j} \geq \frac{(\sum_j s_j)^2}{m}.$$

The mass bounds ensure $m > 0$ when $\varepsilon \leq 1/4$. Combining (26)–(29) yields

$$m \geq \frac{M(1 - 2\varepsilon)^2}{392\varepsilon^2 + \nu(1 + 2\varepsilon)^2 + 2/M}. \quad (30)$$

For $\varepsilon \leq 1/4$, use $(1 - 2\varepsilon)^2 \geq 1/4$, $(1 + 2\varepsilon)^2 \leq 9/4$, and $2/M \leq \nu$ to obtain (7). At $\varepsilon = r^{-1/4}/4$, its denominator is $111r^{-1/2}$. This proves Theorem 1.2. Unlike the preceding counterexample, the coherence step here uses nonnegativity essentially.

5 A probabilistic strengthening

The next noise family has controlled fourth moments and a stronger small-subset singular-value property. All logarithms in this section are natural.

Lemma 5.1 (A noise family with no small nearly dependent subset). *Let $r \geq 4^7$, $N = 100r^3$, and*

$$q_0 = \left\lfloor \frac{r}{64 \log(17N)} \right\rfloor.$$

There are unit vectors $v_1, \dots, v_N \in \mathbb{R}^r$ with mean μ such that

$$\|\mu\|^2 \leq 4/N, \quad (31)$$

$$\frac{1}{N} \sum_t (v_t \cdot y)^4 \leq \frac{19}{r^2} \|y\|^4 \quad (y \in \mathbb{R}^r), \quad (32)$$

$$\left\| \sum_{t \in T} a_t v_t \right\|^2 \geq \frac{1}{64} \sum_{t \in T} a_t^2 \quad (|T| \leq q_0, a_t \in \mathbb{R}). \quad (33)$$

Moreover,

$$q_0 \geq \frac{r}{1152 \log r} \geq 1. \quad (34)$$

Proof. Take independent Gaussian columns $g_t \sim N(0, I_r/r)$, and set $v_t = g_t / \|g_t\|$. These are independent uniform unit vectors. We show that the three events have positive joint probability.

Mean. Independence, symmetry and unit norm give $\mathbb{E} \|N^{-1} \sum_t v_t\|^2 = 1/N$. Markov's inequality bounds the failure probability in (31) by $1/4$.

Fourth moments. For a fixed unit y , put $X_t = (v_t \cdot y)^4$. The spherical moment formulas give

$$0 \leq X_t \leq 1, \quad \mathbb{E} X_t = \frac{3}{r(r+2)} \leq \frac{3}{r^2}, \quad \text{Var } X_t \leq \mathbb{E} X_t^2 = \frac{105}{r(r+2)(r+4)(r+6)} \leq \frac{105}{r^4}.$$

For completeness, the formula $\mathbb{E}(v \cdot y)^{2\ell} = (2\ell - 1)!! / [r(r+2) \cdots (r+2\ell-2)]$ follows by writing a standard Gaussian as its independent radius times its uniform direction, and dividing its one-coordinate moment by its radial moment. These moments follow directly from the one-dimensional Gaussian integral and the chi-square density.

Bernstein's inequality, proved in the form used here in Appendix B, implies

$$\mathbb{P} \left\{ \sum_t X_t \geq \frac{6N}{r^2} \right\} \leq \exp \left(-\frac{9N}{210 + 2r^2} \right) \leq \exp \left(-\frac{N}{24r^2} \right).$$

A $1/4$ -net of the unit sphere has at most 9^r points. The union bound therefore has failure probability at most $\exp(r \log 9 - 100r/24) \leq e^{-r}$. The function $Q(y) = (N^{-1} \sum_t |v_t \cdot y|^4)^{1/4}$ is a seminorm. Its supremum on the unit sphere is at most $4/3$ times its maximum on the net. Raising to the fourth power gives the constant $6(4/3)^4 < 19$ in (32).

Small subsets. Let G be the matrix with columns g_t . For any fixed unit vector a supported on a fixed subset of columns, $r \|Ga\|^2$ has the chi-square distribution with r degrees of freedom. The elementary Chernoff bound in Appendix B gives

$$\mathbb{P}\{\|Ga\| \notin [1/2, 3/2]\} \leq 2e^{-r/16}.$$

For each subset of q columns choose a $1/8$ -net, of size at most 17^q , of its coefficient unit sphere. Union-bounding over all $q \leq q_0$ and at most N^q subsets gives failure probability at most

$$2q_0 \exp(q_0 \log(17N) - r/16) \leq 2r \exp(-3r/64).$$

On the complementary event, the largest singular value of every such submatrix is at most $(3/2)/(1 - 1/8) = 12/7 < 2$. Its smallest singular value is at least $1/2 - (1/8)(12/7) = 2/7 > 1/4$. The case $q = 1$ also gives $\|g_t\| \leq 2$ for every column. Normalizing the columns therefore leaves smallest singular value at least $1/8$, which proves (33).

For $r \geq 4^7$, the sum $1/4 + e^{-r} + 2re^{-3r/64}$ is less than one, so all events can hold simultaneously. Finally, $\log(17N) = \log 1700 + 3 \log r \leq 9 \log r$ for $r \geq 4$. For $r \geq 4^7$ the unfloored quantity defining q_0 is at least two. Replacing its floor by half its value proves (34). $\square$

5.1 Transferring the noise lemma to coreset support

Use the MUB core in dimension r , the N noise vectors from Lemma 5.1, and the rows

$$a_{jt} = N^{-1/4}(u_j, v_t), \quad 1 \leq j \leq M, \quad 1 \leq t \leq N. \quad (35)$$

The original cost satisfies

$$\begin{aligned} f(x, y) &\leq 12 \|x\|^4 + 8 \frac{M}{N} \sum_t (v_t \cdot y)^4 \\ &\leq 12 \|x\|^4 + 114 \|y\|^4, \end{aligned} \quad (36)$$

because $152M/r^2 \leq 114$ for $r \geq 4$. Define $\alpha_{jt} = w_{jt}/N$, $s_j = \sum_t \alpha_{jt}$ and $b_j = \sum_t \alpha_{jt} v_t$; weights may now be arbitrary real numbers. The argument at $y = 0$ is unchanged, so (26) holds.

The mixed coefficient is again $A(x, y) = \sum_j (u_j \cdot x)^3 \langle b_j - \mu, y \rangle$. For unit x, y , the odd parts are bounded by 126ε and 306ε at (x, y) and $(2x, y)$ respectively. Thus

$$|A(x, y)| \leq \frac{306 + 2 \cdot 126}{24} \varepsilon = \frac{93}{4} \varepsilon.$$

Inverting K gives

$$\sum_j \|b_j - \mu\|^2 \leq 961M\varepsilon^2, \quad \sum_j \|b_j\|^2 \leq 1922M\varepsilon^2 + \frac{8M}{N}.$$

Let q_j denote the number of nonzero weights in group j . If $1 \leq q_j \leq q_0$, (33) and Cauchy-Schwarz give $\|b_j\|^2 \geq s_j^2/(64q_j)$. If $q_j > q_0$, use only nonnegativity of squared norm. Both cases imply

$$\|b_j\|^2 \geq \frac{1}{64} \left(\frac{s_j^2}{q_j} - \frac{s_j^2}{q_0} \right), \quad (37)$$

with zero contribution for an empty group. This step remains true for signed coefficients; it is not the coherence argument of Section 4. Summing and using $m = \sum_j q_j$ gives

$$\frac{(\sum_j s_j)^2}{m} \leq 64 \sum_j \|b_j\|^2 + \frac{1}{q_0} \sum_j s_j^2.$$

Substitution proves

$$m \geq \frac{M(1 - 2\varepsilon)^2}{123008\varepsilon^2 + (1 + 2\varepsilon)^2/q_0 + 512/N}.$$

For $\varepsilon \leq 1/4$, and since $N \geq 2048q_0$,

$$m \geq \frac{M}{492032\varepsilon^2 + 10/q_0} \geq \frac{M}{492032\varepsilon^2 + 11520(\log r)/r}.$$

This proves Theorem 1.3.

6 What is resolved, and what is not

6.1 Extension to arbitrary sufficiently large ranks

For a given large integer k , choose the largest power of four r with $2r \leq k + 1$. Then

$$\frac{k+1}{8} < r \leq \frac{k+1}{2}.$$

Append zero coordinates to the constructed rows to obtain dimension $k + 1$. For each query normal from the original $2r$ -dimensional space, append zeros to that normal too. The resulting hyperplane has dimension exactly k and has precisely the same quartic cost. Thus every lower bound transfers. Since $M = \Theta(k^2)$ and $(\log r)/r = O((\log k)/k)$, (9) follows. The constants do not depend on k, ε, n, d .

6.2 Interpretation and remaining scope

The explicit constructions settle the displayed proposed inequality negatively at $p = 4$. The probabilistic strengthening additionally proves (10). The mechanism is that a well-conditioned cubic kernel separates $\Theta(k^2)$ core groups, while the uniform coreset guarantee forces simultaneous approximation of each group’s auxiliary mean. For a suitable noise family this costs order ε^{-2} points per group until the sparse singular-value range is exhausted.

We do not prove a matching upper bound for the expression in (9) outside the range in (10). In particular, (9) saturates near $k^3/\log k$ as ε decreases, whereas the cited general upper bound continues to grow as $\tilde{O}(k^2\varepsilon^{-2})$. Nor do the arguments here establish analogous sharp results for $p \neq 4$. Exact quartic coefficient extraction is essential to the stated proofs. Thus the request to classify the optimal joint function for all fixed $p > 2$ is not fully completed by this manuscript, despite the definitive obstruction to its proposed formula.

7 Reproducible checks and proof audit

The supplied program builds the bases from finite fields, rather than loading precomputed frames. It checks all pairwise differences of the alternating forms and every basis-pair Gram block for $r = 4, 16, 64$. These checks use integer arithmetic. Twenty integer-vector fourth-moment checks are also run in each dimension. They test the implementation; Lemma 2.2 proves the identity for all real vectors.

The program then evaluates the exact mixed-coefficient identities in floating point, verifies the constructive hyperplane witness, and records all four candidate queries. Deliberately half-supported, nonnegative reweightings were used to exercise the witness extraction, not to stand in for the universal lower-bound argument.

r	Rows Mr	Target ε_r	Tested support	Witness relative error
4	48	0.0189959	24	0.746720
16	2304	0.00721688	1152	0.303955
64	135168	0.00260766	67584	0.208059

The largest floating-point extraction residual in the recorded run was below 9×10^{-15} . Full machine-readable results and small matrices are in `results/`. The large random-noise existence argument is proved analytically; the test suite does not claim to instantiate or exhaustively verify that probabilistic family.

The principal audit points are the row-scaling factors, the distinction between entrywise and ordinary kernel powers, normalization of hyperplane normals, and the exact place where nonnegative weights are required. They have been kept explicit throughout. No restriction to oblivious, independent, unbiased or algorithmically generated weights is imposed. The lower bounds cover the best possible support and reweighting after seeing the entire matrix. The package has not been checked in a formal proof assistant or independently peer reviewed.

A A self-contained finite-field construction of the real bases

Complete real MUB families in dimensions that are powers of four are classical; see, for example, Boykin et al. [4] and the Kerdock-code literature [5]. The following construction and proof make our use of them self-contained.

Let $r = 4^h = 2^{m+1}$ with $m = 2h - 1$ odd, let $E = \mathbb{F}_{2^m}$, and let $T : E \rightarrow \mathbb{F}_2$ be the field trace. Thus

$$T(z) = z + z^2 + \cdots + z^{2^{m-1}}, \quad T(z^2) = T(z), \quad T(1) = 1.$$

Consider the $(m+1)$ -dimensional binary vector space $V = \mathbb{F}_2 \oplus E$. For each $a \in E$, define the bilinear form

$$\beta_a((t, x), (s, y)) = T(a^2 xy) + T(ax)T(ay) + tT(ay) + sT(ax). \quad (38)$$

It is symmetric and alternating: on the diagonal, the first two terms cancel because $T((ax)^2) = T(ax) = T(ax)^2$, and the last two cancel.

Lemma A.1. *For distinct $a, b \in E$, the form $\beta_a + \beta_b$ is nondegenerate.*

Proof. Set $c = a + b \neq 0$, $d = a/c$, $X = cx$ and $Y = cy$. Expanding the trace products in characteristic two gives

$$\begin{aligned} (\beta_a + \beta_b)((t, x), (s, y)) &= T(XY) + T(dX)T(Y) + T(X)T(dY) \\ &\quad + T(X)T(Y) + tT(Y) + sT(X). \end{aligned}$$

Suppose (t, x) belongs to the radical. Varying s first gives $T(X) = 0$. The remaining condition for every Y is

$$T([X + T(dX) + t]Y) = 0.$$

The trace pairing is nondegenerate: for nonzero Z , taking $Y = Z^{-1}$ gives $T(ZY) = T(1) = 1$. Consequently $X = T(dX) + t \in \mathbb{F}_2$. Since m is odd and $T(X) = 0$, this forces $X = 0$, and then $t = 0$. Since $c \neq 0$, also $x = 0$. $\square$

Choose any binary coordinate basis $e_1, \dots, e_{m+1}$ of V and define

$$q_a(z) = \sum_{i < j} \beta_a(e_i, e_j) z_i z_j \in \mathbb{F}_2.$$

Its polar form is β_a : $q_a(z + h) + q_a(z) + q_a(h) = \beta_a(z, h)$. For each binary linear functional $\ell : V \rightarrow \mathbb{F}_2$ let the vector indexed by $z \in V$ be

$$u_{a, \ell}(z) = r^{-1/2} (-1)^{q_a(z) + \ell(z)}. \quad (39)$$

For fixed a , the r different functionals give an orthonormal basis, because multiplication by the fixed signs $(-1)^{q_a(z)}$ preserves orthogonality of the binary characters.

If $a \neq b$, an inner product is r^{-1} times a quadratic character sum $S = \sum_z (-1)^{q(z) + \ell(z)}$ whose polar form β is nondegenerate. Its square is

$$S^2 = \sum_{h \in V} (-1)^{q(h) + \ell(h)} \sum_{z \in V} (-1)^{\beta(z, h)} = r.$$

The inner sum vanishes for $h \neq 0$ by nondegeneracy and equals r at zero. Thus different bases in (39) have inner products of absolute value $r^{-1/2}$. The standard coordinate basis is unbiased to all of them, since their entries are $\pm r^{-1/2}$. There are $|E| + 1 = r/2 + 1$ bases in total, as required.

The code encodes field elements as binary polynomials modulo an irreducible polynomial of degree m . It obtains one by exact trial division in the small dimensions used for verification. The integer numerator matrix $\sqrt{r}U$ has entries ± 1 in its flat basis blocks and $\sqrt{r}$ on the diagonal of the coordinate block, permitting exact Gram tests.

B Elementary probability bounds used in the strengthening

We record the estimates to make the positive-probability construction independent of unstated concentration results.

B.1 Finite nets and operator bounds

A maximal η -separated subset of the Euclidean unit sphere is an η -net. The disjoint balls of radius $\eta/2$ around its points lie in a ball of radius $1 + \eta/2$. Comparing volumes gives size at most $(1 + 2/\eta)^r$. If Q is a seminorm and $D = \sup_{\|x\|=1} Q(x)$, then $D \leq \max_{x \text{ in net}} Q(x) + \eta D$. The analogous inequality applies to the operator norm of a linear map. A lower singular-value estimate follows by subtracting the possible operator-norm error $\eta \|G\|_{\text{op}}$ from the net lower bound.

B.2 Bernstein bound for bounded centered variables

Let Z_t be independent with $\mathbb{E}Z_t = 0$, $|Z_t| \leq 1$, and $\sum_t \mathbb{E}Z_t^2 \leq v$. For $0 < \theta < 3$, the exponential series and $j! \geq 2 \cdot 3^{j-2}$ for $j \geq 2$ give

$$\mathbb{E}e^{\theta Z_t} \leq 1 + \frac{\theta^2 \mathbb{E}Z_t^2}{2(1 - \theta/3)} \leq \exp\left(\frac{\theta^2 \mathbb{E}Z_t^2}{2(1 - \theta/3)}\right).$$

Independence and Markov's inequality bound $\mathbb{P}\{\sum Z_t \geq u\}$ by $\exp(-\theta u + v\theta^2/[2(1 - \theta/3)])$. Taking $\theta = u/(v + u/3)$ gives

$$\mathbb{P}\left\{\sum_t Z_t \geq u\right\} \leq \exp\left(-\frac{u^2}{2(v + u/3)}\right).$$

The zero-variance case is immediate. In Lemma 5.1, use $Z_t = X_t - \mathbb{E}X_t$, $v = 105N/r^4$, and $u = 3N/r^2$.

B.3 Gaussian norm tails

For $Z \sim \chi_r^2$, direct Gaussian integration gives $\mathbb{E}e^{tZ} = (1 - 2t)^{-r/2}$ for $t < 1/2$. Optimizing the Chernoff bound at a threshold ar gives

$$\mathbb{P}\{Z \geq ar\} \leq \exp\left[-\frac{r}{2}(a - 1 - \log a)\right] \quad (a > 1),$$

and the same expression bounds $\mathbb{P}\{Z \leq ar\}$ when $0 < a < 1$. For $a = 9/4$ and $a = 1/4$, the exponent coefficient is larger than $1/16$. The union bound therefore gives $\mathbb{P}\{\sqrt{Z/r} \notin [1/2, 3/2]\} \leq 2e^{-r/16}$.

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